Extreme Value Meta-Learning for Few-Shot Open-Set Recognition of Hyperspectral Images

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***Abstract*— Recent advancements in prototype-based few-shot open-set recognition (FSOSR) approaches reject outliers based on the high metric distances from the *known* class prototypes and fail to distinguish spectrally fine-grained land cover outliers. Learning only the Euclidean distance fit spherical distributions ignores the essential distribution parameters such as shift, shape, and scale. The conventional meta-training of FSOSR also ignores the topological consistency of the *known* classes impacting reduced closed and open accuracy in the meta-testing phase. Moreover, the existing hyperspectral outlier detection methods do not provide intuition about the rejected outlier's land cover category. To tackle the aforesaid problems, we introduce extreme value meta-learning (EVML), where we fit Weibull distributions per *known* class based on the limited support-set distances from the respective prototypes. A newly proposed prototypical OpenMax (P-OpenMax) layer leverages these meta-trained Weibull models and calibrates the query distances to reject fine-grained outliers. Then, to learn the topological consistency, we split all the samples in an episode into four parts, including the prototype and its same *known* class queries, other *known* class queries, and the remaining *known–unknown* queries. A novel open quadruplet loss ensures that a prototype's same-class queries reside closer than the other *known*-class and *known–unknown* queries. Finally, we coarse classify the detected outliers into major land cover categories and perform cross-dataset incremental FSOSR to enhance robustness over unknown geographical regions. We validate the efficacy of EVML over four benchmark hyperspectral datasets.**

***Index Terms*— Extreme value theory (EVT), few-shot learning (FSL), hyperspectral classification, open-set recognition (OSR), quadruplet loss.**

I. Introduction

HYPERSPECTRAL image (HSI) classification plays an indispensable role in land cover classification, urban planning, water quality survey, natural resource management, etc. due to its enriched spectral–spatial characteristics. The existing machine learning and deep-learning-based endeavors for HSI classification [1], [2], [3], [4] commonly assume closed-set settings, i.e., HSI pixels should belong to one of the available *known* classes in testing time. Due to the practical limitations of manually inspecting a wide geographical area and high labeling costs by limited available domain experts, many unannotated pixels exist in the benchmark HSI datasets. These unannotated pixels have a minimal spectral difference from the *known* land cover classes. A closed-set trained model is unaware of these spectrally fine-grained outliers and misclassifies them into one of the *known* classes. Furthermore, deploying a closed-set trained model to another geographical area impacts the rise in land cover outliers [5]. Hence, a reliable HSI classifier should learn to reject these outliers and accurately classify the *known* class samples.

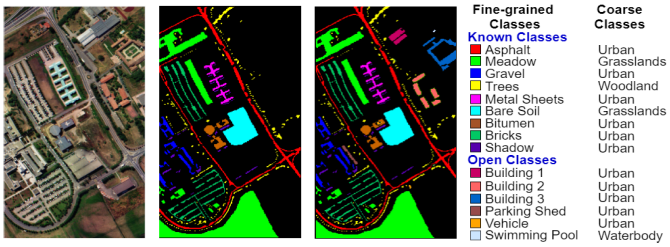

Fig. 1. (Left) Sample RGB bands of the Pavia University dataset. (Middle) Ground truth of nine annotated *known* classes. (Right) Six unknown classes were additionally annotated. A closed-set trained model often misclassifies these unknown samples to one of the *known* classes. We further annotate all the fine-grained HSI classes into a set of coarse-grained categories and train EVML to predict the coarse-grained classes of detected outliers.

The open-set recognition (OSR) [6], [7], [8], [9] methods learn a discriminative boundary for each of the *known* classes to reject an outlier and correctly classify *known* class samples. In literature, training a hyperspectral OSR model and its performance evaluation happens over the same HSI dataset [5], [10], [11], resulting in dataset-dependent multiple model training. However, it limits the deployment of an OSR model over unknown geographical regions. Fig. 1 shows that multiple fine-grained land cover classes can be broadly grouped under a few coarse-grained classes, i.e., "Asphalt," "Brick," "Bitumen," "Buildings," and "Gravel" fall under our annotated coarse-grained class "Urban." There are many remote sensing applications, i.e., urban planning, earthquake and flood damage inspections, satellite image-based navigation map generation, emergency landing of drones, etc., where apart from recognizing these spectrally fine-grained outliers, it becomes useful if

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the outlier's coarse class at least can be predicted. Furthermore, a human annotator can investigate these predicted coarse classes and quickly generate fine-grained labels using domain expertise. Overall, it can drastically reduce the land cover annotation time for the HSI datasets. Nevertheless, there exists no research work in this context. This leads to our first research question "How to infer the coarse class of an outlier from an unknown geographical area over which the model is not trained?"

The existing OSR approaches require ample training data, which is challenging in a few-shot context. MDL4OW [11] tries to develop an OSR model with limited hyperspectral samples, i.e., 20 samples per class, and suffers from model overfitting. Owing to the availability of limited training samples in the HSI datasets, class-imbalance, and the inherent spectrally fine-grained nature of the closed and open-set samples, adapting OSR for HSI becomes challenging. To gain outlier rejection capability from limited data, few-shot OSR (FSOSR) [12], [13] incorporates the few-shot meta-learning or "learning to learn" paradigm in developing an OSR model. FSOSR also tackles the class-imbalance problem as only a few samples are randomly selected per class in episodic meta-training. From the above discussion, using FSOSR for the HSI datasets becomes evident. However, the fundamental question is "How to estimate accurate data distribution from a few training samples in FSOSR?"

Recently, the prototype-based FSOSR approaches [13], [14], [15] have become quite prevalent, where an outlier is recognized based on its very high metric distance from the *known* class prototypes. However, rejecting an outlier based on the Euclidean-distance fit spherical distribution and ignoring the distribution shift, shape, and scale parameters while learning from the heavily skewed HSI datasets become suboptimal for recognizing spectrally fine-grained outliers [see Fig. 2(a)]. Another problem arises from episodic learning in FSOSR. Due to the random sampling of *known* classes in an episode and associated bias in selecting support samples, prototypes are dynamically updated and drift over the metric space. Thus, the true topology of all the *known* classes is not learned in FSOSR. Hence, many fine-grained outliers get misclassified as one of the *known* classes. On the other hand, outliers are *extreme value points* having very low similarity to the *known* class prototypes and reside at the tail of the closed-set distribution. The Weibull analysis can model extreme value distribution (EVD-Type III) [16] and bounds the upper limit of the *known* class distributions having shift, shape, and scale parameters. Leveraging this property, Bendale and Boult [17] and Rudd et al. [18] introduced the Weibull analysis in OSR while proving that the minimum distance between two *known* class samples follows the Weibull distribution. Nevertheless, learning Weibull distributions per *known* class requires robust sample space, as shown in Fig. 2(b), which is critical to obtain in a few-shot setting. This triggers a new research direction, "How to model an accurate constellation of the *known* classes preserving their metric distances along with learning diverse and robust distribution parameters in FSOSR?"

Rejecting fine-grained HSI land cover outliers whose spectral distribution is very similar to the *known* classes is quite difficult in the presence of limited training data. The existing methods enlarge the margin between the *known* samples and outliers by either shrinking the *known* class distribution [19] or bounding the open space [20]. Although Reciprocal Points [20] constrict the open space using the unexplored extra class space of the *known* classes, its performance drops in rejecting the fine-grained outliers from the same dataset [21]. Semantic contrastive mapping (SCM) [22] pulls the *known* samples closer to the prototypes and pushes the outliers away in an open-set domain adaptation context. Triplet loss [23] pulls a positive sample toward the *anchor* and repels a negative sample. P-ODN [14] adapts multiclass triplet thresholding to detect unknowns from the correctly classified training samples. Optimization with triplet loss converges slowly due to considering one negative sample at a time. To speed up, N-pair-mc [24] constructs pairs with one positive and a negative sample from each of the N-1 classes. Lifted Struct [25] constructs a fully connected matrix of the feature pairwise distance, but limited positive samples are used in a batch. The existing FSOSR methods focus only on the relationship between the outliers and the *known* samples but ignore the topology of individual *known* class distributions in the metric space, resulting in reduced closed accuracy. Open accuracy also decreases due to misclassifying outliers to one of the *known* classes.

We propose extreme value meta-learning (EVML), an extreme value theory (EVT) [26]-based meta-learning framework, to calibrate the query distances and separate the fine-grained HSI land cover outliers. For each *known* land cover class, we fit a Weibull distribution based on the support feature distances from their respective prototypes. Then, the query distances of the *known* and *known–unknown* classes are calibrated by a novel prototypical OpenMax (P-OpenMax) layer to ensure *known–unknown* queries reside at the tails of the Weibull distributions or in the open space while *known* queries span only inside the distribution, as shown in Fig. 2(c). Thus, the true *known* land cover class distributions are learned in the metric space using a limited set of training samples. Also, to learn a consistent topology in the metric space, we introduce an open quadruplet loss to simultaneously maintain a consistent metric distance from one *known* class to the set of remaining *known* classes and maximize the metric distance from the outliers. Multiple support-set samples of the *known* classes and queries from the *known* and *known–unknown* classes are simultaneously engaged to optimize this open quadruplet loss. In each episode, we compute multiple *known* class prototypes from the mean of the corresponding support-set samples, which act as *anchors*. Then, we split the queries into three categories. The same *known* class queries of one prototype act as its positive samples. In contrast, the other *known* class queries serve as negative samples of category one, and finally, the *known–unknown* queries are considered as negative samples of category two. The distance between category two negative samples and the *anchor* is much higher than that between the *anchor* and category one negative samples. We estimate the open quadruplet loss by measuring the mean of the quadruplet distances for each *anchor* in an episode. This optimization

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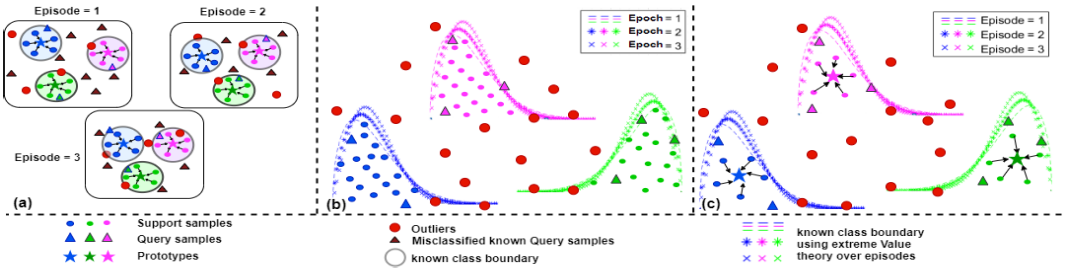

Fig. 2. (a) Conventional FSOSR cannot recognize fine-grained HSI outliers for considering a constant radius from the prototypes. (b) learning Weibull distributions requires large training samples. In contrast, (c) our P-OpenMax layer learns *known* class-specific Weibull distributions centered on the prototypes from limited training samples. Fine-grained outliers are pushed to reside at distribution tails or in the open space due to calibration by P-OpenMax layer.

objective penalizes the fine-grained *known–unknown* outliers to maximize the margin from the closed-set boundary. Our main contributions are as follows.

1) We introduce an EVT-based proposition for FSOSR and EVML, where limited *known* land cover sample distances from their prototypes are used to fit the Weibull distributions, and query distances are calibrated by a novel P-OpenMax layer. EVML better distinguishes fine-grained outliers due to considering robust distribution parameters, shift, shape, and scale and mitigates the need for accurate data distribution learning from a few training samples.
2) Prototype-based open quadruplet loss is proposed for the fine-grained FSOSR task to maintain an accurate constellation of *known* HSI classes in the metric space, along with penalizing the spectrally marginal land cover outliers toward an unbounded open space.
3) To predict the coarse class of a detected outlier over an unknown terrestrial region, we annotate four benchmark HSI datasets into a set of coarse-grained classes by analyzing spectral similarity and then using these coarse classes for meta-learning. Furthermore, we extend our evaluation by incrementally combining multiple HSI datasets to increase the robustness of coarse class prediction.
4) Extensive experiments are performed on four benchmark HSI datasets by assigning spectrally similar land cover classes as closed-open pairs. The model learns to reject the fine-grained outliers, predict their coarse-grained classes, and simultaneously recognize the *known* samples to a set of fine-grained *known* classes.

II. Related Work

A. Few-Shot Meta-Learning

To avoid overfitting while learning from limited training data, few-shot learning (FSL) updates its model parameters using three ways: 1) efficient distance metric learning [27], [28], [29]; 2) using internal or external storage memory [30]; and 3) optimization-based approaches [31] for quickly adapting to a new environment. Majorly, metric learning approaches perform superior to other meta-learning-based approaches. Relation Networks [28] learn the relation score between the support and query features using a relation module. Using attention kernel, Matching Networks [29] learn full context embeddings (FCEs) to relate input with the entire support set. Prototypical Networks [27] compute class centroids of the support set and are computationally fast. Contemporary FSL approaches meta-learn in a closed set setting, thus unable to detect outliers during meta-testing.

B. Open-Set Recognition

During OSR training, a random subset of labeled classes are assigned as *known–unknown* or outlier classes while the rest are designated as *known* categories. The OSR methods can be broadly categorized as: 1) discriminative [32], [33], [34]; 2) generative [35], [36], [37], [38]; and 3) reconstruction-based approaches [39], [40], [41]. Initial deep networks [8] are used to threshold softmax probability scores to reject the outliers. Bendale and Boult [17] opposed the convention as uncertainty is an insufficient condition for outlier rejection and proposed an OpenMax layer. Basically, they applied EVT [26] to fit the Weibull distribution with the distances of the activation vectors from the mean activation vector (MAV) for the correctly classified training samples. Then the softmax probability scores are recomputed based on this trained distribution. Nevertheless, OpenMax fails to recognize fine-grained outliers due to no prior knowledge of open space during training. Furthermore, to hallucinate novel data and learn to reject those pseudo-outliers, G-OpenMax [35] adapts generative adversarial networks (GANs) into the OpenMax framework. An autoencoder trained to reconstruct *known* class data fails to reconstruct unknowns. Using this property, C2AE [40] and CGDL [42] train the autoencoder and recognize outliers at testing time. Perera et al. [43] concatenate the input image with its reconstructed counterpart along the channel dimension. A classifier then distinguishes the input image as *known* or unseen. However, training a deep learning classifier, autoencoder, and GAN requires ample training data that are unsuitable in FSL.

C. Prototype-Based FSOSR Approaches

In each episode of meta-learning in FSOSR, the support set is created from a random collection of *known* classes. In contrast, the query set is constructed with samples from a mixture of *known* and *known–unknown* classes. Generally, prototypical FSOSR approaches compute the *known* class prototypes based

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on the mean of limited support-set samples following the Prototypical Networks [27]. An outlier query becomes better distinguishable due to its high metric distance from the *known* prototypes. PEELER [12] uses the Mahalanobis distance to fit classwise Gaussian distributions but fails to model the data skewness. SnaTCHer [13] rejects outliers producing a high difference between the transformed prototypes and updated prototype set by replacing a query and its predicted prototype. P-ODN [14] learns prototype radius and performs multiclass triplet thresholding to reject outliers. PMAL [15] first mines the high-quality prototypes and then filters to retain a diverse set of prototypes by applying explicit rules. Predominantly, an empirical threshold is used to reject outliers in the above methods. In contrast, OCN [5] uses a variational autoencoder to enrich the *known* class data density and meta-trains a binary classifier to reject the outliers. Even though OCN removes the need for a predefined threshold, focusing on increasing closed data density only creates an imbalance with the open space as ≈10% open accuracy dropped compared with closed accuracy in OCN for the five-shot FSOSR evaluation on the Indian Pines dataset. Also, such a feature hallucination creates spectrally fine-grained novel land cover samples, which are erroneously used to amplify *known* data density. To alleviate this, we use only real *known* samples to fit the closed-set distributions without further feature hallucination and learn to reject outliers as per EVT. Our episodic update of the distributions refines the fine-grained separation of the closed-open boundary.

III. Proposed Methodology

A. Problem Formulation

In FSOSR, a meta-learner learns *known* class distributions from limited training samples using an episodic strategy to reject the outliers and accurately classify the *known* class samples simultaneously. During meta-training, the meta-learner acquires transferable knowledge over a set of *base* classes, and finally, it explores a disjoint set of target classes during meta-testing. To learn to reject outliers in the meta-training phase, a random subset of *base* classes act as *known–unknown* or pseudo-outlier classes representing the open space, with the remainder *base* classes acting as *known* classes. Precisely, a meta-learner explores random $\mathcal{K}$ *known* classes in an episode with m training samples per class, forming the support set $\mathcal{S} = \{(x_i^s, y_i^s)\}_{i=1}^{m\cdot\mathcal{K}}$ and is also termed as $\mathcal{K}$-way m-shot classification. On the other hand, the query set, $\mathcal{Q} = \{(x_j^q, y_j^q)\}_{j=1}^{N\cdot(\mathcal{K}\cup\mathcal{U})}$, is formed with N random samples from each of the $\mathcal{K}$ *known* and $\mathcal{U}$ *known–unknown* classes. As we deal with the spatio-spectral HSI data, $x^s, x^q \in \mathbb{R}^{H\times W\times B}$, where H, W, and B represent the height, width, and spectral bands of the cubic HSI patches, respectively, and y^s and y^q represent the corresponding support- and query set labels, respectively.

B. Overview of the Model Components

In this section, first, we explain the proposed FSOSR approach (B.i), followed by introducing open quadruplet loss (B.ii). Then, we present a novel P-OpenMax layer-based query distance calibration (B.iii) using the proposed EVML approach (B.iv). Finally, we describe the strategy behind the coarse class prediction of the detected outliers by the outlier coarse classifier (B.v).

1) Our FSOSR Approach: Fig. 3 illustrates the proposed EVT-based approach in meta-training *known* class distributions in the realm of FSOSR. We consider R3CBAM from OCN [5] in designing the backbone feature extractor f_φ due to its pioneering work of 3-D HSI feature learning with an attention-aware CBAM3D layer. Although any other backbone can be used as f_φ. R3CBAM extracts the 64-D latent HSI feature vector, and following, we add one dense layer of $\mathcal{K}$ nodes to construct f_φ. We estimate the *known* class prototypes $\mathcal{P}$ by computing the classwise mean of support features $\mathcal{S}_f(\mathcal{K})$. The metric distance of the support features $\mathcal{S}_{\text{dist}}$ from their respective prototypes $\mathcal{P}$ is used to fit $\mathcal{K}$ Weibull models, $\rho_\mathcal{K} = \{\rho_k(\theta_k, \tau_k, \lambda_k)|k = \{1, \ldots, \mathcal{K}\}\}$, parameterized over distribution shift θ, shape τ, and scale λ. On the other hand, we estimate the metric distance, $\mathcal{Q}_{\text{dist}}$, of the combination of *known* and *known–unknown* queries or pseudo-outliers, $\mathcal{Q}_f(\mathcal{K}\cup\mathcal{U})$, from the prototypes $\mathcal{P}$ and followed by a softmax layer produce the closed-set prediction P_{closed}. Then, a novel P-OpenMax layer empowered with the meta-trained closed-set distributions $\rho_\mathcal{K}$ calibrates $\mathcal{Q}_{\text{dist}}$ to produce P_{open} by judiciously unmixing the pseudo-outliers $\mathcal{Q}_f(\mathcal{U})$ from the mixed $\mathcal{Q}_f(\mathcal{K}\cup\mathcal{U})$. During meta-training, P-OpenMax learns transferable knowledge to reject $\mathcal{Q}_f(\mathcal{U})$ by classifying query features $\mathcal{Q}_f(\mathcal{K}\cup\mathcal{U})$ into $\mathcal{K}+1$ classes. To the end, a novel outlier coarse classifier, $\mathcal{O}_\epsilon$, learns to anticipate the coarse-grained classes of the rejected outliers. Finally, during meta-testing, P-OpenMax predicts an unlabeled query as an outlier or *known* sample by leveraging the $\rho_\mathcal{K}$ models. In the case of *known* sample prediction, its class is estimated based on the nearest distance from $\mathcal{P}$, and in the case of outlier prediction, its coarse class is further predicted by $\mathcal{O}_\epsilon$.

2) Open Quadruplet Loss: To minimize the open-space risk, Reciprocal-Point-based classification loss [20] pushes the *known* classes toward the feature space periphery while pulling the outliers toward the center by limiting them in a bounded open space. However, in the case of the HSI datasets, the range of annotated *known* classes is significantly small compared with the set of unannotated diverse land cover classes. Learning to reject the limited set of pseudo-outliers in the meta-training phase of FSOSR becomes suboptimal to bound the open space in the meta-testing phase due to the exposure to novel classes. This shortcoming motivates us to cultivate a closed constellation of *known* classes and penalize the infinite combination of spectro-spatial outliers toward an unbounded open space. Nevertheless, minimizing the interclass variance of *known* classes to form a bounded-closed space can introduce an open challenge of a reduced margin between individual *known* class distributions. Eventually, it impacts reduced closed accuracy due to misclassification of fine-grained *known* class samples. Thus, it further encourages us to formulate a novel optimization function in FSOSR to construct a closed-set topology by maintaining a separation margin between *known* classes. Finally, the closed space is surrounded by unbounded open space. To make it mathematically amenable, we propose

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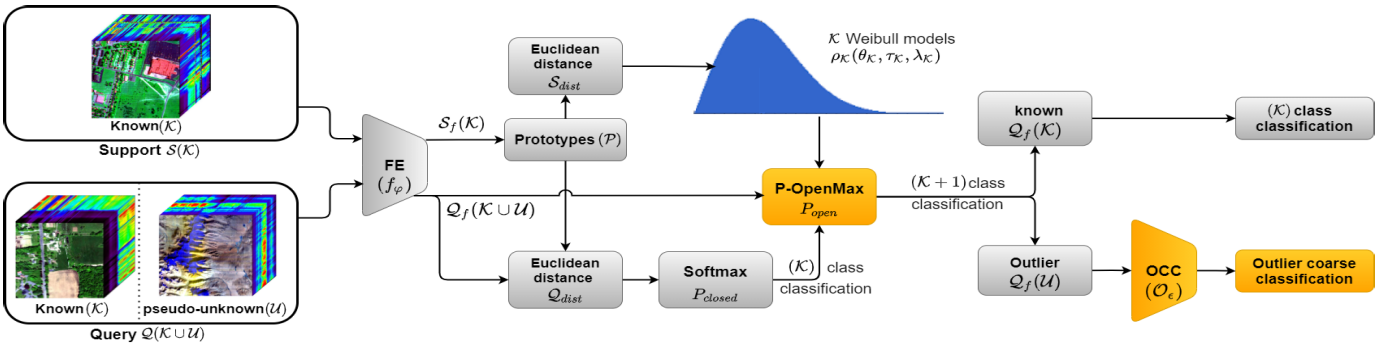

Fig. 3. Architecture of the proposed EVML framework. Once the prototypes $\mathcal{P}$ are measured from the support features, we use the support feature distances $\mathcal{S}_{\text{dist}}$ from $\mathcal{P}$ to fit $\mathcal{K}$ Weibull models $\rho_\mathcal{K}(\theta_\mathcal{K},\tau_\mathcal{K},\lambda_\mathcal{K})$ based on EVT. Then, the query set distance, $\mathcal{Q}_{\text{dist}}$, is calibrated by the P-OpenMax layer. If a query is recognized as *known*, it is further classified into fine-grained $\mathcal{K}$ classes; else, it is classified into coarse-grained classes by outlier coarse classifier, $\mathcal{O}_\epsilon$.

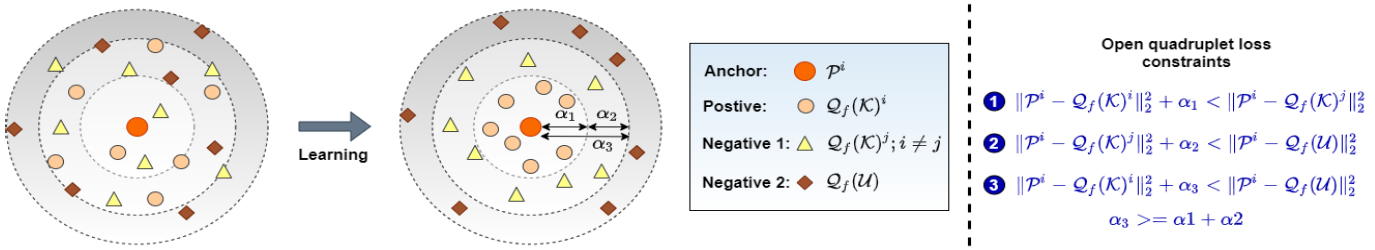

Fig. 4. Illustration of the proposed open quadruplet loss. We enforce three semantic relationships, where: 1) the same *known* class queries $\mathcal{Q}_f(\mathcal{K})^i$ should reside closer to its prototype $\mathcal{P}^i$; 2) other *known* class queries $\mathcal{Q}_f(\mathcal{K})^j$ should maintain at least a separation margin of α_1 from the prototype $\mathcal{P}^i$; and finally 3) the outliers $\mathcal{Q}_f(\mathcal{U})$ should be pushed far away from $\mathcal{P}^i$ at least by a margin of α_3, where $\alpha_3>=\alpha_1+\alpha_2$. [Note: Even though our model is trained with the asymmetric Weibull distribution, we represent query relationships with a prototype using spherical closed-set distributions for visualization simplicity.].

an open quadruplet loss, where each query feature of the ith-*known* class, $\mathcal{Q}_f(\mathcal{K})^i$ with its corresponding prototype $\mathcal{P}^i$, encounters two types of spectrally fine-grained hard negative queries in the meta-training phase. The first one is a different *known* class query, $\mathcal{Q}_f(\mathcal{K})^j$, where $i\neq j$ and the second one is a *known–unknown* query $\mathcal{Q}_f(\mathcal{U})$. We consider three constraints in defining their topological association as follows.

1) Normal positive queries of the ith-*known* class, $\mathcal{Q}_f(\mathcal{K})^i$, are closer to its prototype $\mathcal{P}^i$ than other *known* class queries $\mathcal{Q}_f(\mathcal{K})^j$ by an inter *known* class separation margin of α_1.
2) Negative queries of the jth-*known* class, $\mathcal{Q}_f(\mathcal{K})^j$, are closer to the ith-class prototype $\mathcal{P}^i$ due to the bounded closed space than more negative outlier queries $\mathcal{Q}_f(\mathcal{U})$ by a margin of α_2.
3) Normal positive queries of the ith-*known* class, $\mathcal{Q}_f(\mathcal{K})^i$, are much closer to its own class prototype $\mathcal{P}^i$ than more negative pseudo-unknown queries or outliers $\mathcal{Q}_f(\mathcal{U})$ by a minimum margin of α_3, where $\alpha_3>=\alpha_1+\alpha_2$.

The above constraints help define prototype-based open quadruplet loss, $\mathcal{L}_{O_{\text{quad}}}$ in (1), to ensure that the ith class prototype, $\mathcal{P}^i$, pulls own-class queries $\mathcal{Q}_f(\mathcal{K})^i$ and pushes other *known* class queries $\mathcal{Q}_f(\mathcal{K})^j$ and outliers $\mathcal{Q}_f(\mathcal{U})$ incrementally (see Fig. 4). Our $\mathcal{L}_{O_{\text{quad}}}$ computation differs from the quadruplet loss [44], [45] by considering the prototype as an *anchor* which is a true representative for a *known* class, derived from $\mathcal{S}_f(\mathcal{K})$ and adaptation of it in the FSOSR paradigm

$$\mathcal{L}_{O_{\text{quad}}}=\sum\left[\|\mathcal{P}^i-\mathcal{Q}_f(\mathcal{K})^i\|_2^2-\|\mathcal{P}^i-\mathcal{Q}_f(\mathcal{K})^j\|_2^2+\alpha_1\right]_+$$
$$+\sum\left[\|\mathcal{P}^i-\mathcal{Q}_f(\mathcal{K})^j\|_2^2-\|\mathcal{P}^i-\mathcal{Q}_f(\mathcal{U})\|_2^2+\alpha_2\right]_+$$
$$+\sum\left[\|\mathcal{P}^i-\mathcal{Q}_f(\mathcal{K})^i\|_2^2-\|\mathcal{P}^i-\mathcal{Q}_f(\mathcal{U})\|_2^2+\alpha_3\right]_+ \quad (1)$$

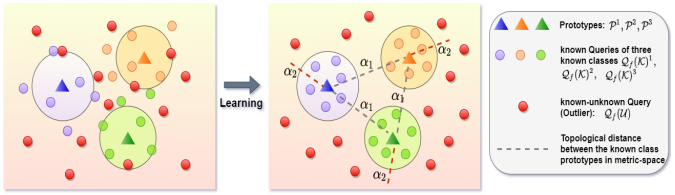

Fig. 5. In a multiclass environment, the proposed open quadruplet loss creates a bounded closed space while maintaining a separation margin α_1 between the *known* classes. It is surrounded by an unbounded open space.

where $i\neq j, \alpha_3>=\alpha_1+\alpha_2, [z]_+=\max(z,0)$, and $\|\cdot\|_2^2$ defines the squared Euclidean distance operator.

The above constraints, while imposed in a multiclass environment, each *known* class maintains a separation margin α_1, forming a topology of discriminative closed space, and the outliers are pushed outward an unbounded open space (see Fig. 5). Thus, the fine-grained separation between the closed and open space is maximized in FSOSR.

3) P-OpenMax Layer: In literature, OpenMax [17] builds a compact abating probability (CAP) [46] model by following the meta-recognition concept [47] and estimates the probability of input being an outlier by normalizing the distance on activation vectors of a pretrained classifier's penultimate layer. However, due to the post-training outlier evaluation by OpenMax, it cannot be incorporated with a meta-learner

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during the meta-training phase, which aims to gain transferable knowledge of outlier rejection over random episodes. Also, different from the feature space learning, where a softmax layer is applied directly on the penultimate layer activation vectors, a meta-learner in the metric space applies softmax on the metric distance of a query from the support-set-derived prototypes [27], [48]. There exists no prior research in meta-learning a closed-set CAP model from limited samples to learn to reject the fine-grained outliers.

In this article, we develop a P-OpenMax layer to bridge this gap by imbibing the meta-recognition idea [47] in FSOSR. Our P-OpenMax layer rejects a query as an outlier by analyzing its distance from the compact abating closed-set distribution, which is meta-trained from limited training samples. Prior meta-recognition works on outlier rejection [17] analyzed the training distribution based on EVT [16] and discovered it follows the Weibull distribution. We also adapt Weibull distributions [49] in fitting each *known* class support feature distances from $\mathcal{P}$. Precisely, we sample $\mathcal{K}$ *known* classes in each episode and meta-train corresponding $\rho_{\mathcal{K}}$ Weibull models parameterized over shift θ, shape τ, and scale λ in Algorithm 1. Each *known* class meta-recognition model dynamically gets updated when its corresponding class is randomly sampled from the *base* classes in an episode. Initially, an untrained feature extractor f_φ generates support features $\mathcal{S}_f$ far from the true prototypes $\mathcal{P}$ in the metric space. Over the episodes, f_φ gradually captures the *known* class distributions while generating $\mathcal{S}_f$ closer to $\mathcal{P}$ or the support feature distances from their respective prototypes $\mathcal{S}_{\text{dist}} \leftarrow \|\mathcal{P}_{\mathcal{K}} - \mathcal{S}_f\|_2^2$ progressively shrink. We use the FitHigh function from libMR [47] to fit $\rho_{\mathcal{K}}$ models on η largest support feature distances per class in each episode. Similar to the notion of the OpenMax layer, which fits meta-recognition models based on the correctly classified data, we update $\rho_{\mathcal{K}}$ models based on the support feature distances only as they are sampled from the *known* classes. Besides, we obtain a closed-set prediction in the metric space for queries $\mathcal{Q}(\mathcal{K} \cup \mathcal{U})$ by applying a softmax layer over the query distance $\mathcal{Q}_{\text{dist}}$ from $\mathcal{P}$, following the Prototypical Networks [27]:

$$P_{\text{closed}}(y = i|\mathcal{Q}_f) = \frac{e^{-\mathcal{Q}_{\text{dist}_i}}}{\sum_{k=1}^{\mathcal{K}} e^{-\mathcal{Q}_{\text{dist}_k}}} \tag{2}$$

where, due to the closed-set assumption, the denominator sums up to 1 as the query with the highest probable $\mathcal{Q}_{\text{dist}}$ will be assigned to the nearest *known* class. Nevertheless, the presence of pseudo-outliers in $\mathcal{Q}_f$ with the associated $\mathcal{Q}_{\text{dist}}$ mandates P_{closed} to be refined considering the open-space risk.

We define the P-OpenMax layer-based open-space probability estimation in Algorithm 2. Let $\mathcal{Q}_{\text{dist}}^{\mathcal{O}}$ represent the outlier distance for all the *known–unknown* samples irrespective of their sampled pseudo-outlier class $\mathcal{Q}(\mathcal{U})$ in an episode. $\mathcal{Q}_{\text{dist}}$ is expected to be refined for all the $\mathcal{K}$ classes selected in an episode, considering the presence of outliers in the open set. Hence, in each episode, we compute the Weibull cumulative distribution function (cdf) probability for unique γ classes, where $\gamma <= |y^s|$, $|y^s|$ denotes the cardinality of y^s. The rejection probability for a query having $\mathcal{Q}_{\text{dist}}$ is calibrated based on the support feature distance fine-tuned $\rho_{\mathcal{K}}$ models.

Algorithm 1 Meta-Learning Steps With P-OpenMax

Meta-Training Phase:
Input: $\mathcal{S}(\mathcal{K})$, $\mathcal{Q}(\mathcal{K} \cup \mathcal{U})$, f_φ, $\mathcal{O}_\epsilon$, η, y^s, y^q, y^c
Output: Updated φ, ϵ parameters
1 Feature extract: $\mathcal{S}_f \leftarrow f_\varphi(\mathcal{S}(\mathcal{K}))$, $\mathcal{Q}_f \leftarrow f_\varphi(\mathcal{Q}(\mathcal{K} \cup \mathcal{U}))$.
2 Compute prototypes $\mathcal{P}$ from *known* class features $\mathcal{S}_f$.
3 Estimate query distances: $\mathcal{Q}_{dist} \leftarrow \|\mathcal{P} - \mathcal{Q}_f\|_2^2$.
4 Estimate closed prediction loss, $\mathcal{L}_{Closed}$ (6) with y^q.
5 Compute open quadruplet loss $\mathcal{L}_{O_{quad}}$ (1)
6 Update $\rho_{\mathcal{K}}$ models based on the Weibull distribution:
EVT Fit $\rho_{\mathcal{K}} = (\theta_{\mathcal{K}}, \tau_{\mathcal{K}}, \lambda_{\mathcal{K}}) = \text{FitHigh}(\|\mathcal{P}_{\mathcal{K}} - \mathcal{S}_f\|_2^2, \eta)$
7 Calibrate $\mathcal{Q}_{dist}$: $P_{open} \leftarrow$ **P-OpenMax**$(\mathcal{Q}_{dist}, \rho_{\mathcal{K}}, y^s)$ and compute open-set prediction loss $\mathcal{L}_{Open}$ (7);
8 Optimize $\mathcal{O}_\epsilon$ by $\mathcal{L}_{OCC}$ (8) for pseudo-outliers to predict the coarse-grained land-cover classes, y^c.
9 Compute $\mathcal{L}_{FE}$ (9) and optimize f_φ parameters.
return Updated f_φ, $\mathcal{O}_\epsilon$ parameters.

Meta-Testing Phase:
Input: $\mathcal{S}(\mathcal{K})$, $\mathcal{Q}(\mathcal{K} \cup \mathcal{U})$, f_φ, $\mathcal{O}_\epsilon$
Output: Predicted classes of the query samples
1 Feature extract: $\mathcal{S}_f \leftarrow f_\varphi(\mathcal{S}(\mathcal{K}))$, $\mathcal{Q}_f \leftarrow f_\varphi(\mathcal{Q}(\mathcal{K} \cup \mathcal{U}))$.
2 Compute prototypes $\mathcal{P}$ from *known* class features $\mathcal{S}_f$.
3 Estimate query distances: $\mathcal{Q}_{dist} \leftarrow \|\mathcal{P} - \mathcal{Q}_f\|_2^2$.
4 Calibrate open space probability, P_{open} by $\rho_{\mathcal{K}}(\theta_{\mathcal{K}}, \tau_{\mathcal{K}}, \lambda_{\mathcal{K}})$ models post fine-tuning with the distances of $\mathcal{S}_f$ from $\mathcal{P}$.
5 **If** $\texttt{argmax}_i P_{open}(y = i|\mathcal{Q}_f) == 0$
Query is detected as outlier and coarse classified by $\mathcal{O}_\epsilon$.
Else
Query is *known*, fine-grained class is $\texttt{argmax}_i P_{open}$
End If
return Predicted classes of the queries.

Algorithm 2 P-OpenMax Layer for Outlier Rejection

Input: Query distance vector : $\mathcal{Q}_{dist}$, $\rho_{\mathcal{K}}(\theta_{\mathcal{K}}, \tau_{\mathcal{K}}, \lambda_{\mathcal{K}})$, y^s
Output: Open space probability P_{open}
1 Let $s(i) = \texttt{argsort}(\mathcal{Q}_{dist})$, and γ is the number of *known* classes to be updated, $\gamma <= |y^s|$.
2 **For** $i = \{1, \ldots, \gamma\}$ **do**
$$w_{s(i)}(\mathcal{Q}_{dist}) = 1 - \frac{\gamma - i}{\gamma} e^{-\left(\frac{\|\mathcal{Q}_{dist} - \theta_{s(i)}\|}{\tau_{s(i)}}\right)^{\lambda_{s(i)}}}$$
End For
3 Revise query distances $\mathcal{Q}_{dist} = \mathcal{Q}_{dist} \circ w(\mathcal{Q}_{dist})$
4 Define $\mathcal{Q}_{dist}^{\mathcal{O}} = \sum_i \mathcal{Q}_{dist} \circ (1 - w_i(\mathcal{Q}_{dist}))$
5 Compute $P_{open}(y = i|\mathcal{Q}_f) = \frac{e^{-1\,\mathcal{Q}_{dist_i}}}{\sum_{k=1}^{\mathcal{K}} e^{-\mathcal{Q}_{dist_k}} + e^{-\mathcal{Q}_{dist}^{\mathcal{O}}}}$
return Open space prediction probability P_{open} of a query.

First, we compute the weight factors for a set of γ *known* classes, where $((\gamma - i)/\gamma)$ is the weight factor with varying $i = \{1, \ldots, \gamma\}$ for γ classes and accordingly scale the Weibull cdf probability score in line 2 of Algorithm 2. The revised query distances for the closed-open set are obtained by elementwise multiplying $\mathcal{Q}_{\text{dist}}$ with the computed Weibull

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scores as

$$\mathcal{Q}_{\text{dist}} = \mathcal{Q}_{\text{dist}} \circ w(\mathcal{Q}_{\text{dist}})$$
$$\mathcal{Q}^{\mathcal{O}}_{\text{dist}} = \sum_i \mathcal{Q}_{\text{dist}} \circ (1 - w_i(\mathcal{Q}_{\text{dist}})) \tag{3}$$

where "$\circ$" defines the elementwise multiplication operator.

The calibrated probability P_{open} for a pseudo-outlier query to reside at $\mathcal{K}$ distribution tails or in the open space is

$$P_{\text{open}}(y = i|\mathcal{Q}_f) = \frac{e^{-\mathbb{1}\,\mathcal{Q}_{\text{dist}_i}}}{\sum_{k=1}^{\mathcal{K}} e^{-\mathcal{Q}_{\text{dist}_k}} + e^{-\mathcal{Q}^{\mathcal{O}}_{\text{dist}}}} \tag{4}$$

where $\mathbb{1} \in \{-1, 1\}$ is an indicator function evaluating to -1 iff $i = 0$ and 1 if $i \neq 0$. Due to the calibration with weighted Weibull cdf probability, P_{closed} for all the $\mathcal{K}$ classes in an episode is also updated in P_{open}. Finally, a query is detected as an outlier when the *known–unknown* class ($y = 0$) obtains the largest probability. Now, we formally define the EVML proposition, which is adapted in the P-OpenMax layer.

4) Extreme Value Meta-Learning: Inspired by the success of EVT [16] on outlier rejection in feature space, we adapt EVT for query distance calibration in the metric space through meta-learning and thereby formally define EVML. Let $(\mathcal{Q}_{\text{dist}_1}, \mathcal{Q}_{\text{dist}_2}, \ldots)$ be a sequence of i.i.d query distances and $\zeta = \max(\mathcal{Q}_{\text{dist}_1}, \mathcal{Q}_{\text{dist}_2}, \ldots)$. If there exists a pair of real numbers (a_n, b_n) such that each $a_n > 0$ and $\lim_{x\to\infty} P(((\zeta - b_n)/a_n) \leq z) = \mathcal{F}(z)$ and $\mathcal{F}$ is a nondegenerate distribution function, then the query distances belong to the Frechet or reversed Weibull family.

5) Outlier Coarse Classifier: The sole objective of $\mathcal{O}_\epsilon$ is to accurately predict the outliers encountered over a different geographical region into a set of coarse-grained classes. In remote sensing, land cover classification requires fine-grained classification of the land cover objects having an infinite combination of spectral–spatial diversities. However, for obvious reasons, most terrestrial objects remain unannotated, causing failure to classify in a fine-grained manner. It motivates us to predict the outliers at least in a coarse-grained level while exposed over a different geographical region or unannotated pixels of the same geographical area. The initial coarse-grained prediction by $\mathcal{O}_\epsilon$ can help a human annotator manually assign the correct fine-grained class due to the reduced search space of the land cover diversities. To comprehend the same, we annotate all the fine-grained classes into a set of dataset-agnostic five major coarse-grained categories (y_c): "Vegetation," "Grasslands," "Woodland," "Urban," and "Waterbody" by exploiting the spectral similarities of the fine-grained HSI classes (see Fig. 10). The annotated coarse-grained classes for four benchmark HSI datasets are shown in Table I, and we show the network architecture of $\mathcal{O}_\epsilon$ in Table II. During meta-training, $\mathcal{K}$ output feature vectors from feature extractor, f_φ, for each pseudo-outlier are processed by $\mathcal{O}_\epsilon$ to meta-learn its coarse category. Finally, $\mathcal{O}_\epsilon$ predicts an unlabeled outlier into one of these coarse classes while exploited over an unseen geographical region during the meta-testing phase.

C. Learning and Inference Protocol

During meta-training, random sampling of episodes produces a disjoint support set, $\mathcal{S}(\mathcal{K})$, and query set, $\mathcal{Q}(\mathcal{K} \cup \mathcal{U})$. We pass them through a feature extractor f_φ to obtain support features, $\mathcal{S}_f$, and query features, $\mathcal{Q}_f$. Then, we measure the *known* class prototypes $\mathcal{P}$ by computing the mean of the support features, $\mathcal{S}_f$, following the Prototypical Networks [27]:

$$\mathcal{P} = \frac{1}{m} \sum_{x_i^s \in \mathcal{S}(\mathcal{K})} f_\varphi\left(x_i^s\right). \tag{5}$$

Initially, we estimate the metric distance of query samples from *known* class prototypes $\mathcal{P}$ or $\mathcal{Q}_{\text{dist}} \leftarrow \|\mathcal{P} - \mathcal{Q}_f\|_2^2$, where $\|\cdot\|_2^2$ represents the squared Euclidean distance. Then $\mathcal{Q}_{\text{dist}}$ is used to successfully minimize the intraknown class variance, maximize the interknown class distance along with pushing away the outliers toward an unbounded open space, and finally, calibrate for the open space risk minimization.

1) Intraknown Class Variance Minimization Loss: We minimize the metric distance of each query feature from the set of *known* class prototypes to learn compact abating and bounded closed-set distribution

$$\mathcal{L}_{\text{Closed}} = \mathbb{E}_{q\in\mathcal{Q}_f(\mathcal{K}\cup\mathcal{U})}\left[-\log \frac{e^{-\mathcal{Q}_{\text{dist}_i}}}{\sum_{k=1}^{\mathcal{K}} e^{-\mathcal{Q}_{\text{dist}_k}}}\right]. \tag{6}$$

2) Interclass Variance Maximization Loss: Furthermore, to increase the separation margin among the *known* classes and push away all the outliers to an unbounded open space surrounding the bounded closed space, we propose to use open quadruplet loss, $\mathcal{L}_{O_{\text{quad}}}$, in (1).

3) Open Space Calibration Loss: The presence of the pseudo-unknown queries to optimize $\mathcal{L}_{\text{Closed}}$ makes the meta-learner vulnerable in rejecting the fine-grained outliers. To learn to reject these fine-grained outliers in each episode, we minimize the negative log probability of a query residing at the *known* class distribution tails and open space by applying the P-OpenMax layer over the query distances

$$\mathcal{L}_{\text{Open}} = \mathbb{E}_{q\in\mathcal{Q}_f(\mathcal{K}\cup\mathcal{U})}\left[-\log \frac{e^{-\mathbb{1}\,\mathcal{Q}_{\text{dist}_i}}}{\sum_{k=1}^{\mathcal{K}} e^{-\mathcal{Q}_{\text{dist}_k}} + e^{-\mathcal{Q}^{\mathcal{O}}_{\text{dist}}}}\right]. \tag{7}$$

At this stage, we go beyond to predict further the coarse classes of the P-OpenMax layer rejected outliers, and henceforth, we meta-train $\mathcal{O}_\epsilon$ with outlier coarse classification loss.

4) Outlier Coarse Classification Loss: Using cross-entropy (CE) loss, $\mathcal{O}_\epsilon$ parameters are optimized over a range of $\mathcal{C}$ coarse classes

$$\mathcal{L}_{\text{OCC}} = \mathbb{E}_{q\in\mathcal{Q}_f(\mathcal{U})}\left[-\sum_{c=1}^{\mathcal{C}} y_c \cdot \log(\mathcal{O}_\epsilon(\mathcal{Q}_{\text{dist}})_c)\right]. \tag{8}$$

a) Total feature extractor loss: Finally, we compute a multitask loss function to optimize f_φ parameters in the following equation:

$$\mathcal{L}_{\text{FE}} = \omega_1 \cdot \mathcal{L}_{\text{Closed}} + \omega_2 \cdot \mathcal{L}_{O_{\text{quad}}} + \omega_3 \cdot \mathcal{L}_{\text{Open}} + \omega_4 \cdot \mathcal{L}_{\text{OCC}} \tag{9}$$

where the $\omega_i, i \in \{1, \ldots, 4\}$ values are the weight factors of different metric loss components.

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TABLE I
ASSIGNMENT OF THE COARSE-GRAINED CLASSES FOR THE PRE-ANNOTATED FINE-GRAINED CLASSES OF THE FOUR BENCHMARK HSI DATASETS

Class No.	Indian Pines		University of Pavia		Salinas		Houston-2013	
	Fine-grained	Coarse-grained	Fine-grained	Coarse-grained	Fine-grained	Coarse-grained	Fine-grained	Coarse-grained
1	Alfalfa	Vegetation	Asphalt	Urban	Brocoli_green_weeds_1	Vegetation	Healthy grass	Grasslands
2	Corn-notill	Vegetation	Meadows	Grasslands	Brocoli_green_weeds_2	Vegetation	Stressed grass	Grasslands
3	Corn-mintill	Vegetation	Gravel	Urban	Fallow	Grasslands	Synthetic grass	Grasslands
4	Corn	Vegetation	Trees	Woodland	Fallow_rough_plow	Grasslands	Trees	woodland
5	Grass-pasture	Grasslands	Painted metal sheets	Urban	Fallow_smooth	Grasslands	Soil	Grasslands
6	Grass-trees	Woodland	Bare Soil	Grasslands	Stubble	Grasslands	Water	Waterbody
7	Grass-pasture-mowed	Grasslands	Bitumen	Urban	Celery	Vegetation	Residential	Urban
8	Hay-windrowed	Grasslands	Self-Blocking Bricks	Urban	Grapes_untrained	Woodland	commercial	Urban
9	Oats	Vegetation	Shadows	Urban	Soil_vinyard_develop	Grasslands	Road	Urban
10	Soybean-notill	Vegetation	Building 1	Urban	Corn_senesced_green_weeds	Vegetation	Highway	Urban
11	Soybean-mintill	Vegetation	Building 2	Urban	Lettuce_romaine_4wk	Grasslands	Railway	Urban
12	Soybean-clean	Vegetation	Building 3	Urban	Lettuce_romaine_5wk	Grasslands	Parking Lot1	Urban
13	Wheat	Vegetation	Parking Shed	Urban	Lettuce_romaine_6wk	Vegetation	Parking Lot2	Urban
14	Woods	Woodland	Vehicle	Urban	Lettuce_romaine_7wk	Vegetation	Tennis court	Urban
15	Buildings-Grass-Trees-Drives	Grasslands	Swimming Pool	Waterbody	Vinyard_untrained	Woodland	Running track	Urban
16	Stone-Steel-Towers	Urban			Vinyard_vertical_trellis	Woodland		

TABLE II
LAYERWISE SUMMARY OF THE OUTLIER COARSE CLASSIFIER

Layer (type)	Output Shape	# Parameter
input_1 (InputLayer)	($\mathcal{K}$)	0
dense_1 (Dense)	(27)	108
dropout_1 (Dropout)	(27)	0
dense_2 (Dense)	(54)	1512
dropout_2 (Dropout)	(54)	0
dense_3 (Dense)	(25)	1375
dropout_3 (Dropout)	(25)	0
dense_4 (Dense)	(5)	130
Total		**3,125**

b) Inference strategy: During meta-testing, we first extract the support and query features through the meta-trained feature extractor, f_φ, and estimate the prototypes from the extracted support features. Then, $\mathcal{Q}_{\text{dist}}$ is measured for a target query from the prototypes and calibrated by the P-OpenMax layer based on the meta-recognition models $\rho_{\mathcal{K}}$. To the end, if the query is recognized as an outlier, its coarse class is predicted by $\mathcal{O}_\epsilon$, else the query is classified as a *known* sample with the highest probable class from P_{open}.

IV. EXPERIMENTS

A. Datasets and Preprocessing

We evaluate our EVML framework over four benchmark HSI datasets.[1] The first one, **Indian Pines (IP)**, was acquired by AVIRIS sensor in North-western Indiana for 145 × 145 spatial resolution with 200 bands after removing the water absorption bands. IP majorly covers 16 fine-grained agriculture and perennial vegetation classes with minor urban objects such as a few low-density houses, a rail line, and highways. The second dataset, **University of Pavia**, was acquired by ROSIS sensor over Pavia, northern Italy, for nine land cover classes with 610 × 610 pixels and 103 spectral bands. Our third dataset for evaluation, **Salinas**, was again captured by AVIRIS sensor over Salinas Valley, California, covering 16 fine-grained classes of vegetables, bare soils, and vineyard fields. Salinas has 512 × 217 spatial dimension and 204 spectral bands after removing 20 bands suffered by water absorption. We consider six more classes from Pavia annotated in [11] for the FSOSR task. Finally, the fourth dataset, **Houston-2013**, has 349 × 1905 pixels with 144 spectral bands and was acquired over the University of Houston for 15 land cover classes.

First, we apply PCA [50] over each dataset to reduce the spectral dimensionality. Thus, the IP, Pavia, and Salinas datasets are reduced to 30 bands, and the Houston dataset is reduced to ten bands, while each dataset is preserving 99% data variance. Then, at each pixel location, we slice cubic patches of dimension (11, 11, ch) using zero padding, where ch indicates the number of spectral bands. We also annotate coarse-grained classes for each dataset, as shown in Table I.

B. Evaluation Metrics

To evaluate the EVML performance, we consider three standard OSR metrics and one classification metric for $\mathcal{O}_\epsilon$. While closed overall accuracy (ClosedOA) represents the percentage of *known* class samples accurately classified, open overall accuracy (OpenOA), on the other hand, evaluates the model performance in recognizing *known* classes and the outliers in (10). Finally, area under receiver operating characteristics curve (AUROC) conveys the outlier detection capability under varying thresholds. Also, we evaluate $\mathcal{O}_\epsilon$ based on the average accuracy

$$\text{Open OA} = \frac{\sum_{k=1}^{\mathcal{K}+1} \text{TP}_k + \text{TN}_k}{\sum_{k=1}^{\mathcal{K}+1} \text{TP}_k + \text{TN}_k + \text{FP}_k + \text{FN}_k} \tag{10}$$

where TP_k, TN_k, FP_k, and FN_k represent the true positive, true negative, false positive, and false negative of the kth *known* class, respectively, and $\mathcal{K}+1$th class denotes all the outliers combined.

C. Experimental Protocol

We evaluate EVML performance based on two settings.

1) Intradataset Evaluation: In this mode of evaluation, the model is evaluated over different datasets independently. Precisely, we randomly assign ten classes as *base* classes for meta-training while the remaining classes are assigned as target classes for meta-testing for an individual dataset. The *base* classes are further split into a spectrally fine-grained set of open and closed classes. Then, we construct a query set with 15 samples per *known* and *known–unknown* class from randomly chosen six *base* classes with random assignment

[1] http://www.ehu.eus/ccwintco/index.php/Hyperspectral Remote Sensing Scenes

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TABLE III

FIVE-WAY ONE-SHOT AND FIVE-SHOT FSOSR PERFORMANCE COMPARISON OF EVML AND SOTA METHODS OVER FOUR HSI DATASETS

Model	Indian Pines			University of Pavia			Salinas			Houston-2013		
	ClosedOA	OpenOA	AUROC	ClosedOA	OpenOA	AUROC	ClosedOA	OpenOA	AUROC	ClosedOA	OpenOA	AUROC
	1-Shot Evaluation											
OpenMax [17]	43.33±0.63	48.54±0.29	44.44±0.33	52.08±0.53	54.61±0.21	52.22±0.35	51.92±0.52	42.50±0.23	50.33±0.34	37.50±0.73	31.33±0.71	36.11±0.33
RDOSR [10]	51.28±0.34	50.13±0.41	47.29±0.23	50.85±0.27	51.68±0.32	55.45±0.42	59.14±0.35	60.19±0.33	54.23±0.21	58.92±0.27	63.53±0.51	61.08±0.34
MDL4OW [11]	46.15±0.21	46.50±0.23	48.66±0.32	56.66±0.22	54.89±0.21	51.11±0.34	58.33±0.21	57.28±0.31	52.77±0.32	41.66±0.43	43.42±0.34	43.88±0.32
PEELER [12]	71.41±0.31	75.45±0.24	71.84±0.21	57.18±0.35	71.55±0.31	52.39±0.18	65.14±0.53	69.63±0.43	57.78±0.33	46.95±0.41	74.34±0.17	53.75±0.32
SnaTCHer [13]	89.33±0.11	81.25±0.23	74.53±0.32	58.50±0.29	75.57±0.34	50.35±0.33	65.94±0.67	78.91±0.28	53.88±0.43	58.73±0.35	77.42±0.33	48.05±0.34
OCN [5]	80.67±0.63	82.72±0.34	75.93±0.32	59.50±0.21	84.77±0.34	89.96±0.43	82.64±0.31	74.27±0.32	81.36±0.21	60.84±0.32	78.93±0.46	74.26±0.17
EVML [Ours]	**92.22±0.17**	**89.83±0.12**	**95.70±0.14**	**71.11±0.08**	**85.52±0.18**	**90.07±0.11**	**85.55±0.07**	**88.55±0.24**	**93.55±0.19**	**72.22±0.16**	**81.25±0.17**	**82.17±0.19**
	5-Shot Evaluation											
OpenMax [17]	51.92±0.34	58.33±0.51	55.62±0.46	69.17±0.32	58.12±0.53	53.36±0.53	69.23±0.25	58.12±0.35	55.44±0.34	41.66±0.24	35.64±0.35	37.84±0.51
RDOSR [10]	55.98±0.51	55.92±0.45	52.38±0.52	64.74±0.45	64.89±0.34	63.94±0.37	67.80±0.35	66.32±0.51	60.28±0.47	68.82±0.52	66.37±0.31	62.56±0.43
MDL4OW [11]	50.76±0.53	46.96±0.35	64.51±0.51	65.33±0.25	62.66±0.19	48.66±0.41	75.01±0.37	62.66±0.34	72.88±0.22	46.66±0.21	44.42±0.23	46.22±0.33
PEELER [12]	82.81±0.39	87.37±0.34	74.19±0.25	60.71±0.23	72.69±0.53	60.36±0.51	74.20±0.52	75.39±0.31	60.39±0.36	57.98±0.47	75.57±0.57	55.47±0.34
SnaTCHer [13]	92.00±0.51	89.42±0.45	76.05±0.55	74.91±0.35	78.10±0.38	54.98±0.29	74.21±0.54	83.15±0.39	72.11±0.43	69.24±0.34	83.29±0.52	49.09±0.45
OCN [5]	94.61±0.38	84.71±0.51	88.40±0.38	71.55±0.32	87.94±0.34	91.90±0.45	84.88±0.37	85.61±0.39	88.52±0.34	71.97±0.29	84.96±0.48	88.50±0.47
EVML [Ours]	**94.94±0.13**	**91.31±0.21**	**96.23±0.17**	**86.67±0.14**	**89.28±0.21**	**92.03±0.12**	**94.44±0.17**	**91.21±0.17**	**96.29±0.09**	**86.54±0.16**	**87.38±0.28**	**94.18±0.23**

of three classes as *known* and the remaining as *known–unknown*. Besides, we construct the support set with m disjoint samples of the three *known* query classes. To optimize f_φ and $\mathcal{O}_\epsilon$, we use the Adam optimizer [51] with a learning rate of 0.0001 and 0.001, respectively. Also, the value of η is assigned as m in Algorithm 1 (refer Table VII). The margins of open-quadruplet loss [see (1)] are set as $\alpha 1 = 0.5$, $\alpha 2 = 0.5$, and $\alpha 3 = 2$ after validation in Table VIII. We assign the weight factors in (9), $\omega_1 = 0.4$, $\omega_2 = 0.2$, $\omega_3 = 0.2$, and $\omega_4 = 0.2$ after cross-validation in Table IX. During meta-testing, we randomly sample episodes 500 times from the set of target classes as per the literature [5] and note the mean ± standard deviation values for a reliable prediction in Table III.

2) Cross-Dataset Incremental Learning: We meta-train our model over a single dataset or combination of datasets and thereby perform meta-testing over other unseen datasets to evaluate the coarse-class recognition performance of $\mathcal{O}_\epsilon$. EVML gets incrementally meta-trained over the *base* classes of the mentioned HSI datasets and gains transferable knowledge of outlier coarse classes spread over unknown distributed geographical regions. All other experimental setups are the same as the above experimental protocol.

D. Comparison With Other FSOSR Methods

1) Overall Performance: To verify the efficacy of our approach, we evaluate the proposed EVML framework against the state-of-the-art (SOTA) OSR and FSOSR methods over the four benchmark HSI datasets. The results for the one-shot and five-shot FSOSR comparison are shown in Table III. Thanks to EVML's compact *known* class distribution learning ability from the limited training samples, it shows superior fine-grained outlier rejection ability in FSOSR compared with the other SOTA methods over the HSI datasets.

OpenMax [17] was developed to reject the outliers during the testing phase based on learning from the large-scale training datasets. Consequently, it produces inferior results in recognizing the HSI outliers in a few-shot context. To adapt OpenMax in FSOSR, we fit the query distances from prototypes in *known* class Weibull distributions and rejected the outliers only in the meta-testing phase. MDL4OW [11] fits the reconstruction error in EVT and got overfitted due to not applying the meta-learning paradigm while learning from limited training data. We use the reconstruction distance of 3-D HSI patches to fit *known* class prototypes and adapt MDL4OW in FSOSR. An outlier is rejected due to having a high reconstruction distance from all the *known* class prototypes. Similarly, we incorporate RDOSR [10] with Prototypical Networks [27] to extend its HSI outlier rejection ability in a few-shot context. PEELER [12] and SnaTCHer [13] were introduced initially to tackle the FSOSR problem, and we readily evaluated them over the HSI datasets. Our baseline FSOSR method, OCN [5], evaluated their architecture over the HSI datasets only. We use a common feature extractor, R3CBAM, from OCN to extract HSI features in evaluating OpenMax, PEELER, and SnaTCHer. For the one-shot evaluation, EVML beats the next best FSOSR alternative over a wide margin in Table III. We observe a significant 19.76% AUROC gain over IP, 11.61% ClosedOA increase over Pavia, 12.19% AUROC boost over Salinas, and an 11.38% OpenOA rise over the Houston-2013 dataset. Similarly, EVML exhibits quite a dominating performance over the five-shot FSOSR task. EVML beats the following best alternatives by 7.83% AUROC over IP, 11.76% ClosedOA over Pavia, 9.56% ClosedOA, and 7.77% AUROC over Salinas, and a significant 14.57% ClosedOA for the Houston-2013 dataset. In terms of OpenOA, EVML beats the benchmark OCN by 7.1% for one-shot and 6.60% for five-shot evaluation over the Indian Pines dataset.

2) Qualitative Performance: The generated classification maps by different FSOSR methods are compared in Fig. 6. We annotate the fine-grained open classes in "White" color. In the case of the Salinas dataset, PEELER misclassifies many HSI pixels from the *known* class "Vinyard_untrained" and assigns them as outliers. SnaTCHer also predicts *known* "Vinyard_untrained" pixels to outliers. None of the existing methods could recognize *known* pixels from the "Vinyard_untrained" class as it is spectrally similar to the annotated open classes. On the other hand, EVML recognized all the closed-open classes with high accuracy, including the "Vinyard_untrained" class. In the case of the Pavia dataset, we observe utter outlier rejection performance by PEELER and SnaTCHer. EVML recognizes the majority of open-class pixels and obtains very high closed-set recognition performance on the Pavia dataset. Similarly, EVML shows a stellar performance over IP, even recognizing the queries from the smallest open class, "Grass-pasture-mowed." For the Houston-2013 dataset, the maximum number of pseudo-unknown pixels from the "Running track" class are recognized by EVML.

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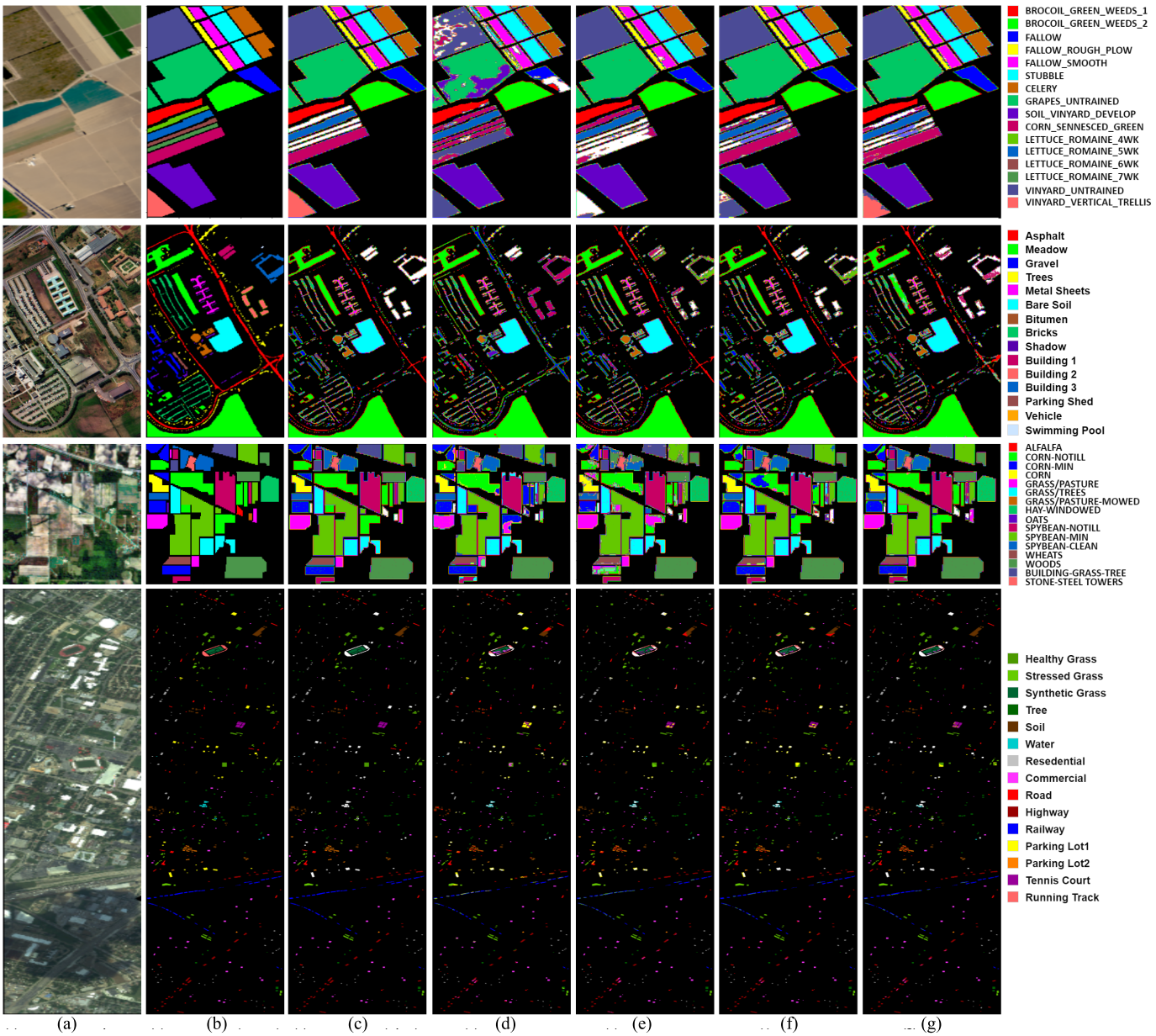

Fig. 6. Comparison of land cover classification maps for the five-shot FSOSR by SOTA methodologies (d) PEELER, (e) SnaTCHer, (f) OCN, and proposed (g) EVML over (Row 1) Salinas (Row 2) University of Pavia, (Row 3) Indian Pines, and (Row 4) Houston-2013 datasets. (a) True color composite for each dataset, and the corresponding annotated ground truth for closed and open sets are shown in (b) and (c), respectively, with open classes marked in "White."

We anticipate this dominant performance of EVML is due to the fine-grained outlier separation by the open quadruplet loss coupled with the closed-open calibration by the P-OpenMax layer. EVML also achieves the maximum area under the ROC curve in Fig. 7.

E. Further Analysis

1) Effect of Different Optimization Functions: We compare the convergence stability of the proposed open quadruplet loss against CE, triplet [23], and Reciprocal-Point-based classification loss [20] in Fig. 8. CE only minimizes intraclass variance by assigning each *known* sample to its closed-set class and all the pseudo-outliers to a single *known–unknown* class. Fine-grained outliers remain unseparated from *known*-class boundaries in Fig. 9(a), impacting reduced OpenOA and AUROC in Table IV. To optimize with triplet loss, we form a positive pair between a *known* class prototype and its queries, another negative pair between that prototype, and a combination of all other *known* class queries and *known–unknown* queries. Nevertheless, triplet loss also produces reduced OSR performance in Table IV as negative pair samples are mixed with other *known* class queries and outliers [see Fig. 9(b)]. Reciprocal Points create a bounded open space while spreading the *known* queries toward the feature space periphery in Fig. 9(c). Due to high spectral similarity, some fine-grained outliers cause reduced discriminability but overall improve over triplet loss by 3.95% OpenOA and 7.9% AUROC on the IP dataset. Finally, our open quadruplet loss helps maintain a semantic topology between *known* classes and spreading outliers to surround unbounded open space [see Fig. 9(d)], securing a

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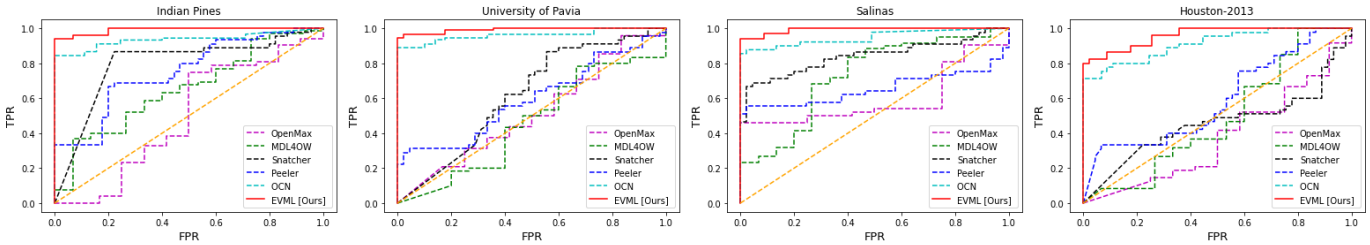

Fig. 7. Comparison of the AUROC plots by different methods for the five-shot FSOSR task over four benchmark HSI datasets. EVML (plotted in "Red" color) demonstrates the best true-positive rate (TPR) performance over other SOTA methods for a low false-positive rate (FPR) value.

TABLE IV

PERFORMANCE COMPARISON OF DIFFERENT LOSS FUNCTIONS TO OPTIMIZE EVML FOR THE FIVE-SHOT FSOSR TASK

Optimization Objective	Indian Pines			University of Pavia			Salinas			Houston-2013		
	ClosedOA	OpenOA	AUROC	ClosedOA	OpenOA	AUROC	ClosedOA	OpenOA	AUROC	ClosedOA	OpenOA	AUROC
Cross Entrpy	92.67±0.31	74.34± 0.27	78.44±0.19	82.22±0.28	77.11±0.23	78.19±0.41	91.72±0.27	76.67±0.34	79.77±0.31	80.41±0.18	78.90±0.28	80.25±0.24
Triplet [23]	86.16±0.54	77.18±0.39	75.72±0.25	83.20±0.32	80.65±0.25	80.53±0.17	85.29±0.23	79.88±0.35	82.63±0.18	75.98±0.26	72.46±0.31	74.31±0.34
Reciprocal [20]	94.44±0.22	81.14±0.26	83.62±0.19	83.89±0.14	86.83±0.27	85.90±0.32	92.78±0.24	82.11±0.36	86.45±0.24	82.35±0.25	83.37±0.21	85.95±0.17
Open quadruplet	**94.94±0.13**	**91.31±0.21**	**96.23±0.17**	**86.67±0.14**	**89.28±0.21**	**92.03±0.12**	**94.44±0.17**	**91.21±0.17**	**96.29±0.09**	**86.54±0.16**	**87.38±0.28**	**94.18±0.23**

TABLE V

ANALYSIS ON THE NUMBER OF META-RECOGNITION MODELS IN EVML TO BOUND THE CLOSED SPACE FOR THE FIVE-SHOT FSOSR TASK

Model	Indian Pines			University of Pavia			Salinas			Houston-2013		
	ClosedOA	OpenOA	AUROC	ClosedOA	OpenOA	AUROC	ClosedOA	OpenOA	AUROC	ClosedOA	OpenOA	AUROC
1-Weibull	91.33±0.12	78.11±0.23	80.01±0.27	85.90±0.07	80.97±0.26	79.27±0.31	92.19±0.24	78.89±0.21	81.79±0.17	85.27±0.23	75.29±0.18	77.21±0.24
$\mathcal{K}$-Weibulls	**94.94±0.13**	**91.31±0.21**	**96.23±0.17**	**86.67±0.14**	**89.28±0.21**	**92.03±0.12**	**94.44±0.17**	**91.21±0.17**	**96.29±0.09**	**86.54±0.16**	**87.38±0.28**	**94.18±0.23**

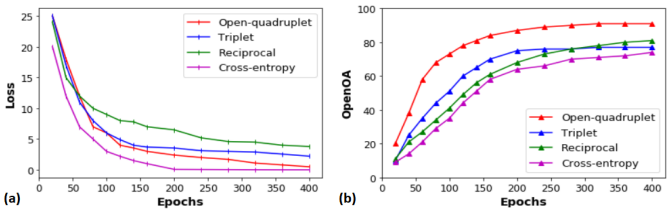

Fig. 8. (a) Model convergence comparison by optimization with different loss functions. (b) Even though CE decays fastest, it shows poor OpenOA performance.

significant 10.17% OpenOA and 12.61% AUROC gain over IP, 6.13% AUROC over Pavia, 9.09% OpenOA over Salinas, and a massive 8.23% AUROC rise over Houston compared with the next best optimization function alternative.

2) Number of Meta-Recognition Models: We perform an ablation experiment to estimate the closed-set distribution using a single Weibull model versus a set of $\mathcal{K}$-Weibull models in an episode in Table V. Basically, a single Weibull model-based experiment estimates the entire closed-set distribution considering support feature distances of all *known* classes from their respective prototypes; outliers are rejected based on its query feature distance calibration by the P-OpenMax layer from this single Weibull model. On the other hand, the $\mathcal{K}$-Weibull experiment creates per *known* class Weibull model, and the outliers are rejected based on their query feature distance from the individual of these $\mathcal{K}$ meta-recognition models. We observe that $\mathcal{K}$-Weibull-based EVML beats the 1-Weibull model by 13.2%, 8.31%, 12.31%, and 12% OpenOA over Indian Pines, Pavia, Salinas, and Houston datasets, respectively. AUROC values also increased proportionately. It is due to the fact that having $\mathcal{K}$ meta-recognition models, the outliers are repelled from a set of *known* classes compared with a single closed-set model in 1-Weibull, thus creating more separation margin between the closed-open data distributions. Therefore, it boosts ClosedOA due to maximizing the interclass variance.

3) Selection of the Coarse-Grained Classes: The fine-grained HSI land cover classes having similar spectral signatures reside in affinity in the feature space and form coarse-grained clusters. The t-SNE visualizations of such coarse-grained classes for the four benchmark HSI datasets are shown in Fig. 10. Based on this interclass spectral similarity, we have further annotated all the fine-grained HSI classes into five major coarse-grained categories, namely, "Vegetation," "Grasslands," "Woodland," "Urban," and "Waterbody" for the incremental FSOSR learning task. During meta-training, $\mathcal{O}_\epsilon$ learns to predict the coarse-grained class of the rejected outliers. Finally, in meta-testing, a query from an unknown geographical region is initially predicted as an outlier by the P-OpenMax layer. Followed by, $\mathcal{O}_\epsilon$ predicts its coarse-grained class. Overall, we find four coarse-grained classes of IP, namely, "vegetation," "grassland," "woodland," and "urban," as IP was predominantly captured over the agriculture region. Similarly, the University of Pavia has four major coarse-grained classes, "grassland," "woodland," "waterbody," and "urban," where many fine-grained classes are clustered in the "urban" category. Salinas has a vast area covered with "grassland," "woodland," and "vegetation" classes. The Houston-2013 dataset majorly contains urban samples with minor "waterbody," "grassland," and "woodland" classes. We define the fine-grained annotations to "coarse-grained" class assignments for the above-mentioned four HSI datasets in Table I.

4) HSI Cross-Dataset Incremental Learning: Our HSI dataset incremental meta-learning strategy toward a robust FSOSR helps in learning finer closed-set distribution boundaries in the metric space while reducing the degree of openness uncertainty for a trained model exploited over a different

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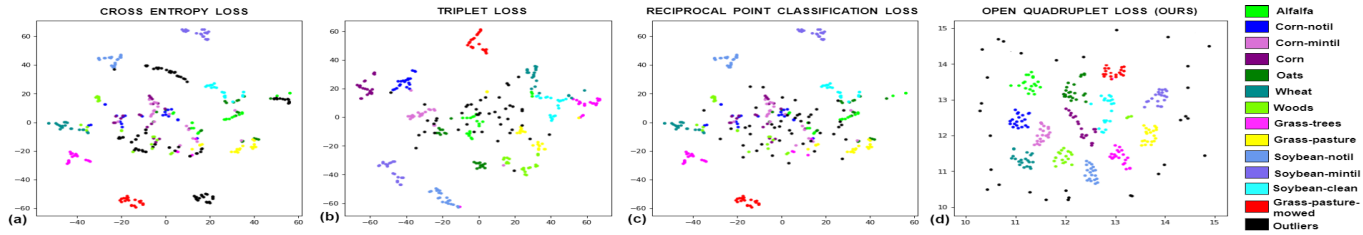

Fig. 9. t-SNE visualization of the metric space due to optimizing EVML over the Indian Pines dataset. It can be observed that only the open-quadruplet loss separates the fine-grained HSI outliers ("Black" dots) by penalizing them in an unbounded open space. (a) cross-entropy loss, (b) triplet loss, (c) reciprocal point classification loss, and proposed (d) open quadruplet loss.

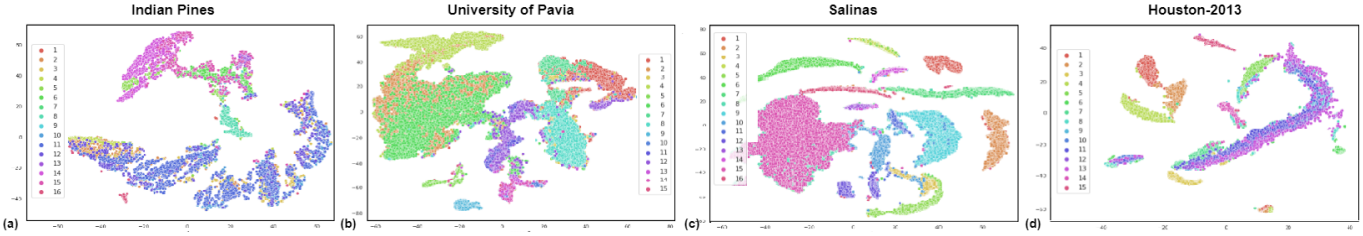

Fig. 10. t-SNE visualization of land cover classes for four benchmark HSI datasets, (a) Indian Pines, (b) University of Pavia, (c) Salinas, and (d) Houston-2013, is plotted. Several fine-grained land cover classes in each dataset can be broadly clustered into a coarse-grained class based on similar spectral responses.

TABLE VI

HSI CROSS-DATASET INCREMENTAL LEARNING FOR A ROBUST FINE-GRAINED FSOSR TASK BY THE PROPOSED EVML FRAMEWORK

Meta-training Datasets	Shot	Meta-testing Dataset											
		Indian Pines (IP)			Salinas (Sal)			University of Pavia (Pav)			Houston-2013 (Hous)		
		Fine-grained		Coarse-grained	Fine-grained		Coarse-grained	Fine-grained		Coarse-grained	Fine-grained		Coarse-grained
		ClosedOA	OpenOA	Accuarcy	ClosedOA	OpenOA	Accuarcy	ClosedOA	OpenOA	Accuarcy	ClosedOA	OpenOA	Accuarcy
IP	1	**92.22±0.17**	**89.83±0.12**	**96.07±0.16**	78.23±0.31	74.05±0.28	74.66±0.37	62.22±0.19	78.14±0.32	76.88±0.42	67.33±0.21	76.27±0.41	72.20±0.27
IP+Sal	1	90.97±0.27	82.34±0.29	94.95±0.16	**84.99±0.18**	**77.02±0.24**	**76.11±0.47**	62.02±0.31	79.27±0.19	78.66±0.17	67.44±0.38	77.23±0.29	72.56±0.17
IP+Sal+Pav	1	89.27±0.21	80.89±0.34	93.34±0.27	82.53±0.36	75.06±0.18	75.34±0.27	**69.88±0.55**	**84.15±0.25**	**82.22±0.18**	68.54±0.34	77.89±0.43	73.29±0.39
IP+Sal+Pav+Hous	1	87.85±0.26	79.28±0.14	91.37±0.55	81.86±0.23	73.67±0.24	73.34±0.18	67.23±0.34	83.85±0.36	80.98±0.37	**70.55±0.32**	**79.38±0.12**	**75.55±0.47**
IP	5	**94.94±0.13**	**91.31±0.21**	**97.25±0.17**	89.32±0.26	82.55±0.18	81.09±0.34	81.77±0.13	84.22±0.46	87.79±0.38	83.66±0.34	84.66±0.21	83.33±0.38
IP+Sal	5	92.93±0.23	85.81±0.13	96.12±0.31	**93.93±0.17**	**85.34±0.21**	**85.55±0.36**	83.33±0.47	85.77±0.39	88.87±0.54	84.28±0.23	85.38±0.39	85.23±0.27
IP+Sal+Pav	5	91.87±0.32	84.39±0.16	95.16±0.22	91.79±0.29	84.38±0.36	83.43±0.27	**86.44±0.15**	**88.83±0.31**	**92.02±0.37**	84.88±0.24	86.55±0.42	85.55±0.18
IP+Sal+Pav+Hous	5	89.73±0.55	82.59±0.23	94.55±0.16	90.42±0.34	82.78±0.19	81.67±0.28	85.28±0.34	86.39±0.14	90.73±0.29	**85.55±0.31**	**86.55±0.45**	**87.66±0.35**

geographical region. Simultaneously, it helps in maturing $\mathcal{O}_\epsilon$ to learn to predict the coarse-grained class of a rejected outlier from an unknown geographical region. In Table VI, we show the dataset incremental learning performance of EVML by successively meta-training it over different HSI datasets and performing meta-testing over the target classes. First, we meta-train EVML over IP to learn to reject the fine-grained agricultural outliers. Then, we successively select different datasets, Salinas, Pavia, and Houston, and randomly assign a set of *base* classes to fine-tune EVML and perform meta-testing over their target classes. Unlike IP and Salinas, which primarily covers different agriculture regions, Pavia and Houston are enriched with urban classes. The same EVML model is meta-trained over heterogeneous geographical regions to learn to reject fine-grained outliers over widely diversified agriculture and urban regions. When we meta-train EVML for multiple *base* classes from different datasets, CloseOA, and coarse-grained accuracy decays compared with the baseline EVML model trained over IP as a query can reside in other spectrally similar *known* classes. OpenOA also degrades slightly as fine-grained outliers get misclassified to one of the *known* classes. For one-shot evaluation, the coarse-grained classification accuracy of $\mathcal{O}_\epsilon$ becomes 91.37% over IP target classes after meta-training over *base* classes from all the four datasets. Increasing the support size to 5, we get 3.18% better accuracy. Again, when EVML is trained over all the four datasets, including Houston, we observe better meta-testing accuracy over Houston than the model meta-trained without Houston, as EVML could not discover significant discriminative urban features in the Houston area. Similar meta-recognition performance is observed for all the experiments in Table VI.

TABLE VII

IMPACT OF THE TAIL SIZE VARIATION OVER THE INDIAN PINES DATASET

Tail Size (η), m Shots	ClosedOA	OpenOA	AUROC	Coarse Acc
$\eta = 1, m = 1$	92.22	89.83	95.70	96.07
$\eta = 1, m = 5$	93.10	90.38	95.08	96.19
$\eta = 2, m = 5$	93.44	89.31	95.44	96.31
$\eta = 3, m = 5$	93.78	90.22	95.84	96.59
$\eta = 4, m = 5$	94.02	90.58	96.05	96.95
$\eta = 5, m = 5$	94.94	91.31	96.23	97.25
$\eta = 10, m = 10$	95.91	92.44	96.76	97.45
$\eta = 15, m = 15$	96.17	92.55	96.49	97.56

5) Impact of Tail Length for Query Distance Calibration: In EVML, the outliers with extreme query distances from the prototype should reside at the tail of a Weibull distribution. Each *known* class distribution gets meta-trained with η largest support feature distances from their corresponding prototype in an episode, or η can also be considered as tail length. However, as EVML aims to learn distributions from limited support samples, the upper bound of tail length becomes the support-set size (m) in an episode. We performed an ablation study in Table VII over the Indian Pines dataset by varying this tail length, η, along with the number of support features

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TABLE VIII

FIVE-SHOT FSOSR PERFORMANCE EVALUATION OVER IP BY VARYING THE MARGINS IN OPEN-QUADRUPLET LOSS

Margins	Condition	ClosedOA	OpenOA	AUROC	Coarse Acc
$\alpha1 = 0.5, \alpha2 = 0.5, \alpha3 = 1$		92.81	90.91	94.81	95.73
$\alpha1 = 0.5, \alpha2 = 0.5, \alpha3 = 2$	$\alpha3 \geq \alpha1 + \alpha2$	**94.44**	91.31	**96.23**	**97.25**
$\alpha1 = 0.5, \alpha2 = 0.5, \alpha3 = 2.5$	$\alpha1 = \alpha2$	94.25	**91.79**	95.36	97.11
$\alpha1 = 0.2, \alpha2 = 0.8, \alpha3 = 2$	$\alpha3 > \alpha1 + \alpha2$	92.89	89.71	94.11	94.89
$\alpha1 = 0.8, \alpha2 = 0.2, \alpha3 = 2$	$\alpha1 \neq \alpha2$	93.65	88.39	93.86	94.22

TABLE IX

WEIGHT COEFFICIENTS OF DIFFERENT LOSS COMPONENTS IN OPTIMIZING TOTAL FEATURE EXTRACTOR LOSS OVER THE INDIAN PINES DATASET

$\{\omega_1, \omega_2, \omega_3, \omega_4\}$ values	ClosedOA	OpenOA	AUROC	Coarse Acc
0.1, 0.4, 0.4, 0.1	88.32	90.85	94.44	94.81
0.4, 0.1, 0.4, 0.1	94.54	86.39	88.45	91.59
0.4, 0.4, 0.1, 0.1	93.11	89.48	93.27	93.41
0.5, 0.2, 0.2, 0.1	**95.14**	90.11	95.95	94.79
0.4, 0.2, 0.2, 0.2	94.94	**91.31**	**96.23**	**97.25**

(m shots) in an episode to see the improvement in the closed and open accuracies. Keeping a constant m-shot while reducing η, the values of ClosedOA, OpenOA, and AUROC decrease progressively. On the other hand, while increasing the tail length and the number of shots, all these accuracy metrics steadily increase. This phenomenon is due to the fact that while fitting a Weibull distribution with a more number of the largest support feature distances, the distributions effectively learn a robust standard deviation for all the *known* samples from their prototype, thus increasing ClosedOA. Fine-grained outliers also get recognized better due to penalization from the largest distant *known* class support samples, which increase OpenOA and AUROC.

6) Impact of Margins in Open-Quadruplet Loss: In designing the open quadruplet loss [see (1)], we define three margins: 1) α_1 controls the maximum range of query distances from a corresponding prototype; 2) α_2 controls the separation of other *known* class queries; and 3) α_3 controls the minimum lower distance between the outliers from the *known* class prototypes. Table VIII shows EVML's FSOSR performance by changing these margins.

In the case of ($\alpha3 > \alpha1 + \alpha2$) with $\alpha1 = \alpha2$ and $\alpha3 = 2$, we see an increased OpenOA as EVML better distinguishes the fine-grained outliers compared with the case of ($\alpha3 = \alpha1 + \alpha2 = 1$). This is because the higher the α_3 value, the annulus margin between the inner bounded closed space and the outer unbounded open space increases in the metric space leading to a better outlier detection ability of the model. Furthermore, increasing α_3 to 2.5, we observe no improvement as outliers were already well separated at $\alpha_3 = 2$. Again, keeping $\alpha3 = 2$, we vary $\alpha1$ and $\alpha2$ and observe all the metric values decrease. Especially when $\alpha1 < \alpha2$, EVML fails to estimate an accurate density of the *known* classes. However, when $\alpha1 > \alpha2$, the *known* class samples get sparsely spread around their prototypes, and other *known* samples get squeezed up between the boundaries between $\alpha1$ and $\alpha2$, leading to some of it escaping into the open space. It results in poor ClosedOA and OpenOA. The optimum choice for all the margins is $\alpha1 = \alpha2 = 0.5$ and $\alpha3 = 2$.

7) Weight Coefficients of Loss Components: In (9), we introduce four weight coefficients in defining the total feature extractor loss, and we perform the five-shot FSOSR over the Indian Pines dataset to cross-validate their values in Table IX. When $\mathcal{L}_{\text{Closed}}$ coefficient ω_1 is very low compared with ω_2 and ω_3, limited *known* class queries are pulled toward their prototype, and mostly they got recognized as outliers. Hence, OpenOA, AUROC, and Coarse Acc increased over ClosedOA. Again, assigning a very low value to ω_2 compared with ω_1 causes fine-grained outliers to remain unrecognized, impacting OpenOA and AUROC performance. Outliers are better distinguishable once we assign ω_2 an increased value. However, giving an equal priority to ω_2 and ω_3, with their summation having a marginal difference to ω_1 boosts all the metric values. It happens because $\mathcal{L}_{O_{\text{quad}}}$ detects fine-grained outliers, and $\mathcal{L}_{\text{Open}}$ performs hand-on-hand to calibrate the open space. Finally, we observe a significant Coarse Acc gain when ω_4 is increased to 0.2. Even though ClosedOA slightly reduces due to lower ω_1, it helps achieve the highest score over the remaining three metrics.

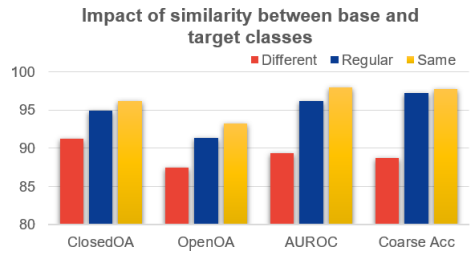

Fig. 11. Similarity between the *base* and target classes of the Indian Pines dataset helps achieve better closed and OSR performance.

8) Consequence of the Base and Target Classes Similarity: We consider three scenarios in sampling classes from the Indian Pines dataset according to the coarse-grained annotations in Table I.

a) Same: Ten *base* classes are formed by randomly selecting six vegetation classes from a total of nine, two grassland classes out of four, one woodland class from two options, and one and only urban class, "Stone-Steel-Towers." The remaining six classes are assigned to target classes. The "Same" class sampling strategy ensures that the meta-training and meta-testing classes have equivalent coarse categories.

b) Different: During meta-training, we select all the nine vegetation classes with one urban class to construct a set of ten *base* classes, whereas target classes for meta-testing are formed with the four grassland classes and two woodland classes.

c) Regular: The classes are randomly sampled irrespective of the *base* and target class similarity. We follow this class-splitting protocol in evaluating all the methods in Table III.

In Fig. 11, we observe very high ClosedOA and AUROC performances following the "same" strategy. It happens as EVML already acquires significant transferable knowledge to distinguish the *known* and unknown samples from similar *base* classes. However, approximately a 3% performance drop happened for ClosedOA and OpenOA, and a significant 9% Coarse Acc was reduced to deal with the "Different" category target classes compared with the "Regular" option. Also, to enable "Different" class-splitting, we had to fine-tune with

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TABLE X
MODEL COMPLEXITY COMPARISON OVER IP (FIVE-WAY FIVE-SHOT)

Model	GFLOPS	Parameters	AUROC
OpenMax [17]	0.045	38.5K	55.62
RDOSR [10]	2.1	400K	52.38
MDL4OW [11]	2.3	643K	64.51
PEELER [12]	0.045	40K	74.19
SnaTCHer [13]	0.045	40K	76.05
OCN [5]	0.045	40.5K	88.40
EVML (Ours)	0.047	41K	**96.23**

TABLE XI
FSOSR PERFORMANCE COMPARISON BY VARYING BACKBONES FOR FIVE-WAY FIVE-SHOT EVALUATION OVER THE INDIAN PINES DATASET

Backbone	ClosedOA	OpenOA	AUROC	Coarse Acc
EVML + SSRN [4]	40.82	48.46	50.71	65.19
EVML + 3D-CNN [52]	37.28	45.96	46.79	61.77
EVML + D-Res-3D-CNN [52]	77.51	69.37	70.14	82.96
EVML + HybridSN [3]	41.67	50.38	52.01	68.37
EVML + R3CBAM [5]	**94.94**	**91.31**	**96.23**	**97.25**

limited samples of the target classes as outlier coarse classifier, $\mathcal{O}_\epsilon$, was unaware of the disjoint coarse labels of the target classes.

9) Memory Footprint: EVML has two learnable network components, feature extractor f_φ and outlier coarse classifier $\mathcal{O}_\epsilon$. f_φ has 38 309 parameters and $\mathcal{O}_\epsilon$ has 3125 parameters (see Table II). A few other additional parameters are shift θ, shape τ, and scale λ for each *known* class Weibull distributions. In Table X, we compare the space complexity of different methods against EVML. We consider R3CBAM from OCN [5] in evaluating OpenMax, PEELER, and SnaTCHer for the HSI datasets. Even though EVML has a slightly higher computational complexity than [5], [12], [13], and [17], it secures the highest AUROC.

10) Influence of the Backbone on Feature Extractor: In Table XI, we inspect the influence of different backbones on the feature extractor. SSRN [4] uses serially connected two spectral-residual blocks followed by two spatial-residual blocks. Even though the residual connections help combat the gradient vanishing problem over vanilla 3D-CNN [52], irrelevant features get extracted due to a lack of attention-aware learning. Without having any residual connection and attended feature extraction, HybridSN [3] combines conv3D and conv2D layers and secures poor performance in a few-shot context. Having residual 3D-CNN layers to learn spectro-spatial HSI features jointly, D-Res-3D-CNN [52] achieves better performance. Furthermore, R3CBAM [5] incorporates attention map learning after each convolution block of D-Res-3D-CNN. Overall, EVML with R3CBAM as the backbone feature extractor extracts the salient HSI features, which is essential in learning from the limited labeled data.

V. CONCLUSION

In this article, we propose a novel FSOSR method, EVML, for open-set remote sensing image classification in the low-data regime, by exploring the coarse-grained and fine-grained associations among the class semantics. Precisely, EVML seeks to meta-learn the characteristics of the closed-set classes from the available limited labeled samples. On the other hand, we introduce a novel P-OpenMax layer that calibrates the metric distance of a query with respect to the support samples for spectrally fine-grained outlier rejection using a novel open quadruplet loss. Since the existing datasets do not have both coarse and fine-level annotations available, we introduce the coarse-grained labels for the land cover classes of four benchmark HSI datasets. Finally, we introduce a novel experimental scenario for cross-dataset incremental learning, enabling EVML to be applied over novel geographical areas and predict the coarse class of the detected outliers. Experimentally, EVML beats the previous literature consistently. The fine-grained outlier rejection performance can be further boosted by considering more support feature distances to fit per-class Weibull models due to better approximating the closed-set data density. Our future work will consider incorporating the notion of open-domain generalization from few-labeled training samples, which is an important task, yet an unexplored problem in the context of remote sensing.

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